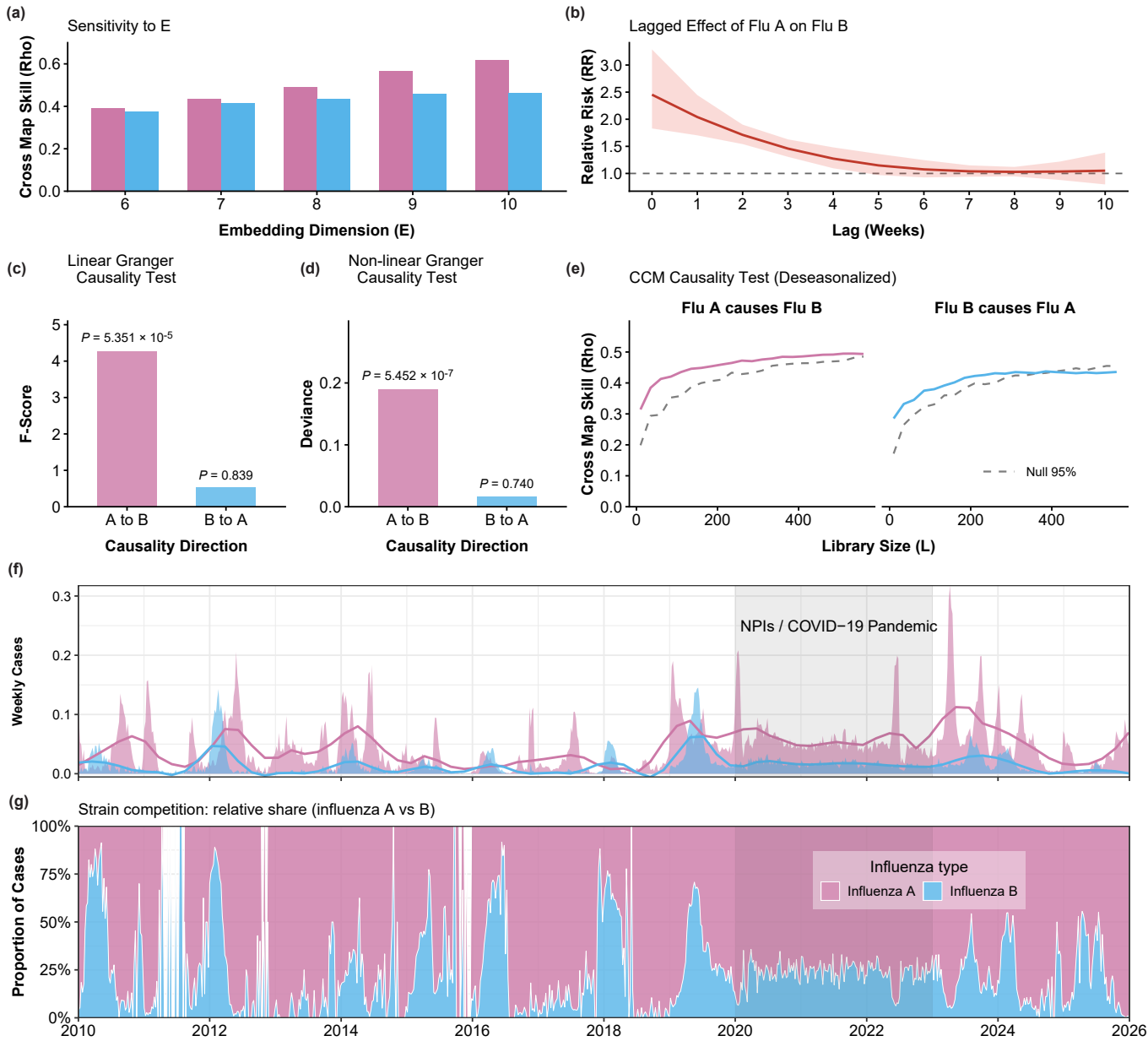

(a)
Sensitivity to E
Cross Map Skill (Rho)
0.0
0.2
0.4
0.6
6
7
8
9
10
Embedding Dimension (E)
(b)
Lagged Effect of Flu A on Flu B
Relative Risk (RR)
1.0
1.5
2.0
2.5
3.0
0
1
2
3
4
5
6
7
8
9
10
Lag (Weeks)
(c)
Linear Granger Causality Test
F-Score
P = 5.351 × 10-5
P = 0.839
A to B
B to A
Causality Direction
(d)
Non-linear Granger Causality Test
Deviance
P = 5.452 × 10-7
P = 0.740
A to B
B to A
Causality Direction
(e)
CCM Causality Test (Deseasonalized)
Flu A causes Flu B
Flu B causes Flu A
Cross Map Skill (Rho)
Null 95%
0
200
400
Library Size (L)
(f)
Weekly Cases
NPIs / COVID−19 Pandemic
(g)
Strain competition: relative share (influenza A vs B)
Proportion of Cases
100%
75%
50%
25%
0%
Influenza type
Influenza A
Influenza B
2010
2012
2014
2016
2018
2020
2022
2024
2026